

Bogmayer Automated Analytics Platform



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Abstract

The Bogmayer Automated Analytics Platform is a dashboard that increases data accessibility and data literacy for the employees of Bogmayer Ltd. beverage investment company. The purpose of this program is to consolidate, analyze, and broadcast sales and operational data reports for the sales team and higher officials to understand and make informed business decisions to catalyze production. It is a temporary solution until the company grows, to ensure that data is being evaluated and saving money before hiring a full-time analyst.

This platform operates by automatically pulling reports from multiple data sources, storing them securely in a backend database, performing exploratory data analysis and visualization, generating summaries of the data, and automatically emailing a report and the raw data files to pertinent employees in the company. Bogmayer Ltd. is looking to capitalize on expansion opportunities and increase sales revenue to optimize the growth of the company.

Data-driven decision making is at the forefront of our future, and with the rise of artificial intelligence, this dashboard will be the first step toward building a systematic way to expand Bogmayer Ltd. I look to build upon the website further to optimize the output and continue making tasks simpler for the employees at Bogmayer Ltd.

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Chapter 1

Introduction

Project Rationale

As technology continues to evolve, businesses are rushing to utilize new platforms to get more information to make better strategic decisions. The ability to understand and respond to the needs of customers is essential to outperform competitors. Many companies like Bogmayer Ltd. are investing great sums of money to gain an advantage on their data driven decision making (DDDM) by implementing new methods of analysis technologies.

I am currently employed by Bogmayer Ltd. as a Data Analyst for the semester. My job is to investigate why specific sales phenomena occur, send reports, and help other employees become more data literate. I was inspired to make this tool for Bogmayer Ltd. for my capstone project because it integrates the skills that I have learned at Hendrix College with my work as a Data Analyst to create an important tool that will be useful.

Bogmayer Ltd. has recently struggled with their strategic decision making by failing to capture critical information from their data platforms. The company does not have an analytics team that looks at the data platforms that they are paying for, so a large amount of the data was left unanalyzed. As they expand, this will become a major issue, causing the company to miss out on key opportunities and lose money.

Goals of the Project

The main goal of this project is to design an automated data analysis dashboard that consolidates data from multiple platforms into a singular reporting system. It looks to improve data driven decision making by creating visualizations and providing insights that highlight successes and improvements for the company. This allows Bogmayer Ltd.'s leadership to make impactful decisions quickly and well.

Figuring out how to successfully make this platform was a time-consuming process that integrated different concepts that I have learned in my Computer Science courses at Hendrix College and on my own time. I also learned a lot of different techniques and aspects of development. This platform combines aspects of data science, software development, data analytics, and research into a consolidated environment to help Bogmayer Ltd. understand their data in more depth.

The Bogmayer Automated Analysis Dashboard is made from HTML/CSS, Python/Flask, and Selenium. This online dashboard autonomously downloads reports from Bogmayer Ltd's different data platforms, conducts Exploratory Data Analysis, creates visualizations, and provides a write-up of the findings. The dashboards are autonomously sent out each week to inform the company's management for them to get a better understanding of what is happening in the back end of the company.

Significance for the Project

This platform will help simplify the complexities and confusions of running a company by providing helpful information about sales, inventory, tasting events, and potential target locations. These are the first steps towards the company making better business decisions that are based on concrete evidence and reasoning.

Because Bogmayer Ltd.'s sales team has access to all of the sales data in their email, there will be an increase in efficiency of the employees. With this information, our sales team will increase their productivity by determining which retailers are out of stock and need to be visited. With Bogmayer Ltd's executives having information in front of them, they will be lead to make better decisions to grow the company. Being provided concrete data, visualizations, and insights about what is occuring

within the company will lead to better, strategic decision making for optimization in production and sales.

Chapter 2

Background

Business Intelligence and Strategy

With globalization and advances in technology, companies today operate in fast-paced environments that require quick and accurate decision-making. To be competitive in modern markets, organizations have limited room for error when analyzing large volumes of data and information. [1]

To combat this challenge, advancing technologies are taking a major role in producing faster, more accurate, data-driven decisions. Business Intelligence (BI) has revolutionized the way that companies operate and handle big data to progress.

Over the past two decades, BI tools have advanced reporting by integrating quantitative results, visualizations, and trends into consolidated dashboards that help provide insightful information. Even though Artificial Intelligence (AI) has only recently been introduced into the business world, it has made drastic impacts on real-time data analysis that has accelerated the growth of many companies globally.

Bogmayer Ltd. is looking to join this trend by implementing AI and Machine Learning (ML) to help analyze data and perform basic, redundant tasks for strategic growth and to improve efficiency in the office.

Analytics

Analytics can be described as a statistical way to assess data, understand patterns, insights, and trends that can be impactful in informed decision-making. It provides

businesses with the ability to understand why certain phenomena are happening in the company, forecast future trends, and optimize operations.

Statistical analysis can be classified into three different branches: descriptive, predictive, and prescriptive. Descriptive statistics involves looking at past data to gain insight about what strategies are working and what can be improved. Predictive statistics look into the future using stochastic methods to forecast trends and guide decision-making. Prescriptive analytics diagnoses problems by examining both past and future data to identify patterns. [14]

This framework is the foundation of modern Decision Support Systems (DSS). Scott-Morton defined DSS as “interactive computer-based systems, which help decision-makers utilize data and models to solve unstructured problems” [6]. These systems help bridge the gap between human judgment.

Today, data is one of a company’s most valuable assets. Scott-Morton divides data into three different forms: structured data, unstructured data, and semi-structured data. Structured data is stored in databases in a tabular format, like sales or inventory records. Unstructured data is qualitative, such as customer feedback, while semi-structured data combines both forms.

Companies pay massive amounts of money for data service platforms and analyst roles because they understand its value. In my project, I alleviate the stressful decision-making process for Bogmayer Ltd. by providing automated insights into data from their various subsidiaries.

Importance of Data-Driven Decision Making

In current markets, Data-Driven Decision Making (DDM) is an integral part of business success. Being able to evaluate risks, identify growth opportunities, and measure performance can greatly influence success. This leads to smarter, more informed decisions, higher productivity, and overall growth. [2]

To optimize profits and production, integrating a culture that relies on DDDM promotes company success. By creating an automated analysis dashboard, everyone in the company—from the sales team to the CEO—will be more informed about the company’s performance. This transparency allows employees to take on more tasks and make timely decisions without spending time looking for information.

Tools and Packages Used

The Bogmayer Automated Data Analysis Dashboard combines a variety of technologies for its functionality. The tools and packages used include the following:

- **HTML/CSS/JavaScript:** Used to design an intuitive, user-friendly front-end interface.
- **Python & Flask [9] :** Serve as the back-end framework for data processing, routing, and system logic.
- **Selenium: [13]** Automates the extraction and downloading of data from web-based platforms.
- **Pandas [10] & YData [15] :** Provide data cleaning, manipulation, exploratory analysis, and visualization.
- **Gmail API [5]:** Enables automated report distribution to management on a scheduled basis.
- **pdfkit [11] and PyPDF2 [12] :** Converts python findings and ydata html file into a pdf report.

Combined, these tools create a functional analytics pipeline that automatically generates reports from different data platforms and distributes them to management on a weekly basis.

Business Needs of Bogmayer Ltd.

As a beverage investment company, Bogmayer Ltd. focuses on potential expansion locations, prior sales, inventory, and tastings data to foster growth. There are three different tiers that Bogmayer Ltd. works with: supplier, distributor, and retailer.

The suppliers produce all of the products and ship them to distributors. The distributors then deliver them nationwide to retailers, who sell directly to consumers. Bogmayer has access to data from all three sectors, helping to analyze the supply

chain and flow of goods throughout the Southeast. Understanding these relationships is key to identifying growth areas.

As the company expands and gains more partnerships, the data becomes increasingly complex. Each partner uses different systems, making the information unconsolidated and difficult to interpret. Without an analytics team, much of this information is ignored, leading to suboptimal decisions.

The Bogmayer Analytics Platform resolves this issue by operating as a centralized data platform that automates reports, visualizations, and insights on all available data. Comparing data from multiple platforms reveals discrepancies and performance deltas, helping management make informed, evidence-based decisions.

The project was successful because I was able to:

- Create data pipelines from multiple different platforms to my site
- Conduct automated analysis on the data that is pulled
- Create insightful visualizations
- Create template reports that are automatically sent out each week

Stages of Development

When I started my project, I partitioned it into three different phases: Platform Development, Analytics, and Reporting. This allowed me to focus on specific aspects of it and progressively improve my project while making sure I always had some part of my platform always working.

Stages 1: Platform Development

In this stage, I focused on getting the functionality of a tangible platform that is able to conduct analytics when data is uploaded. This is the concrete foundation for my project and consists of two different components: Front End and Back End.

Analytics

In this stage, I focused on applying analytical techniques to extract information from the data that was uploaded to the platform. Analytics are an integral aspect of gaining meaning and value from data. I have three different stages of analytics: Exploratory Data Analysis, Visualizations, and Insights. These three stages will provide insightful information to Bogmayer Ltd. by showing different patterns, statistics, and anomalies in the data.

Reporting

In this stage, I focused on broadcasting the results that I found in my analytics phase to the executives of Bogmayer Ltd. Before I completed my analytics phase, I started this phase and got an automated email system working. When I finished the analytics, I adjusted my reports to be the specific reports that I wanted to send out. The ability to display your results for other people to understand is crucial because the results are not valuable if they are not interpretable or understandable.

Chapter 3

Body

Problem Presented

The current issue in Bogmayer Ltd. is that the company does not currently have an analytics team to leverage data to help make insight driven decisions. Large amounts of the data is not being evaluated and the decisions that are being made may not be optimal because they are made based upon a feeling rather than trends and facts. Implementing data-driven decision making into Bogmayer Ltd's executive decision making will result in optimized profits, expansion in size and locations, and more strategic choices. Sub-optimal decision making will hinder the growth of the company. The guidelines of this project were to create a computer program that successfully enhances the data literacy of the employees of Bogmayer Ltd. and helps the executives make informed decisions in a three month time frame using the skills that I currently know and have learned.

Brainstorming

Before development started, I strategically planned how I could resolve the issue with a computational platform. I spent a long time gathering ideas, talking to the executives at Bogmayer, and exploring what software other companies use that could guide my work. Although there were no platforms that were exactly what I was envisioning, I was inspired by multiple different features that I could implement into my work. Important elements to include were data frames, visualizations, statistical analysis, and written insights to help identify KPI's (Key Performance Indicators)

and growth for the company. After gathering the ideas, I researched what languages and tools I would need to reach functionality. Then I sketched basic wireframes to try to consolidate my thoughts into a singular, achievable design. Below is my initial time line (Table 3.1) created at the start of the project to ensure that I completed everything on time:

Step	Task	Deadline
1	Set up GitHub, HTML foundations of website, be able to upload data from computer	September 5th
2	Basic Auto EDA upon selected data	September 19th
3	Auto Visualization	October 3rd
4	Generate insights about the data and visualizations	October 10th
5	Generate a template PDF report with findings and email weekly	October 17th
6	Create data pipelines from data sources in addition to manual uploading	October 31st
7	Look further into analytics (clustering and time series comparisons)	November 14th
8	Add final touches and complete the project	November 28th

Table 3.1: Project Timeline and Milestones

Front End

To start off the project, I created a basic HTML/CSS framework for what I wanted the dashboard to look like. I have prior experience making websites, so this was a simple start to the project. I played around and set up the Front End to look like how I wanted the dashboard to look at the end. I loaded in sample data and images to try to imagine how I can optimize the production and make my brainstorming come to life. After this, JavaScript was added to load a csv file into the dashboard and display it in a scroll-able manner.

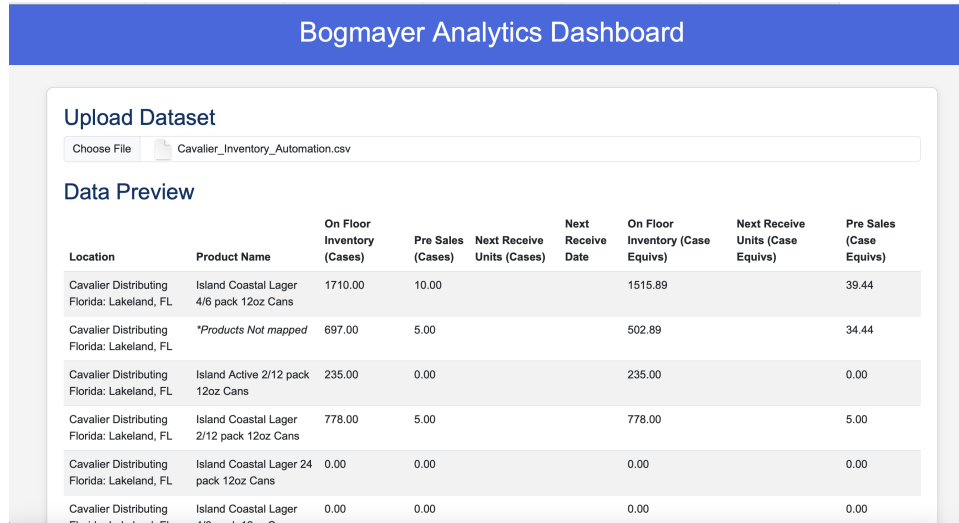


Figure 3.1: Front-End Image

The interface displays:

- Key Performance Indicators such as total sales, revenue growth, and product performance.
- Conduct automated analysis on the data that is pulled
- Navigation options for viewing historical data, current summaries, and exportable reports.

Designing the front end was a linear functionality that came with minimal errors or conflict, but was more a part of my design phase. The colors and formatting of the dashboard represent the company's color scheme and values.

Back End

The back end of my platform is set up using Python (Flask) [9] to store the navigation routes of my things. This was an important step for my project because it stores the data that we can examine and manipulate them to find results. With the data in the back end, Pandas [10], Python packages, and external scripts are able to be used on it. I had not had much experience with back end development, but learned

about routes and how integral it is to retrieve the analysis, visualizations, and insights functional.

EDA

Exploratory Data Analysis (EDA) is an integral part of business intelligence and data driven decision making. My primary goal of this part of the dashboard was to create statistics and information that everyone in our company is able to interpret and have a better understanding of Bogmayer's Ltd's production and sales. Initially, I wanted to use an API key from large Artificial Intelligence companies, but I ran into many complications, including: implementation into my code, price, potential mistakes, and data security. After running into these complications, I switched routes and decided to run an Auto-EDA package on my dataframe out of Pandas. To do this, I had to set up the back end with Flask to handle the csv when it is uploaded to the front end. This was an unfamiliar task for me, but after reading online resources, I successfully added this to my project. Doing this allows me to run Pandas and have control of manipulating the content of the uploaded data. After setting up the back end, two different subsections of EDA were created: a section of basic descriptive statistics and a section of an Auto EDA package analysis.

1. Basic Descriptive statistics:

In this section, I needed to get some form of EDA completed by my designated schedule while my back end was still in progress, so my dashboard finds basic descriptive statistics. I ran a script to display the column name, datatype, example cell, minimum, maximum, and the number of missing cells for each column.

Basic Data Analysis

Rows: 6, Columns: 9

Column	Type (guess)	Missing	Example	Stats
Location	string	0	Cavalier Distributing Florida: Lakeland, FL	-
Product Name	string	0	Island Coastal Lager 4/6 pack 12oz Cans	-
On Floor Inventory (Cases)	number	0	1710.00	min: 0, max: 1710, mean: 570.00
Pre Sales (Cases)	number	0	10.00	min: 0, max: 10, mean: 3.33
Next Receive Units (Cases)	number	6		min: infinity, max: -infinity, mean: NaN
Next Receive Date	number	6		min: infinity, max: -infinity, mean: NaN
On Floor Inventory (Case Equivs)	number	0	1515.89	min: 0, max: 1515.89, mean: 505.30
Next Receive Units (Case Equivs)	number	6		min: infinity, max: -infinity, mean: NaN
Pre Sales (Case Equivs)	number	0	39.44	min: 0, max: 39.44, mean: 13.15

Uploading to backend for deeper analytics...

Figure 3.2: Basic Descriptive Statistics Output

2. YData:

Progressing upon the previous stage, I tried different Auto EDA packages on the Pandas data frame in my back end. I initially ran SweetViz [3] that produced my best results, but it would crash my program when I tried to download it. Then I switched to use YData [15] that has the same functionality as SweetViz, but uses less memory, so it works better to run in my dashboard. This was successful in printing in-depth insights that will help the company.

Automated EDA Report

Below is the automatically generated analysis report:

Automated Profiling Report

Overview

Variables

Correlations

Missing values

Sample

Overview

Brought to you by YData

Overview

Alerts11

Reproduction

Dataset statistics

Number of variables	9
Number of observations	6
Missing cells	18
Missing cells (%)	33.3%
Duplicate rows	0
Duplicate rows (%)	0.0%
Total size in memory	564.0 B
Average record size in memory	94.0 B

Variable types

Categorical	5
Text	1
Unsupported	3

Figure 3.3: YData Analysis Output

Visualizations

In addition to the visualizations produced by YData, I created my own graphics that are specified to Bogmayer Ltd.’s needs. I originally did not plan on doing this because I thought the API key would handle the graphics automatically, but this did not work, so I rerouted my plan. The visualizations are created using Pandas to read the file in the back-end and then Matplotlib [7] to create the plots. The dashboard is designed to produce two specific visualizations for saved tables that I run through it every week, but I included fallback visualizations if the columns are not recognized as the ”standard” report that I send out every week.

Figure 1

For the first graphic, if the data that is being interpreted has values named “Charleston” in a column named “Location”, then it produces a bar graph of the inventory at the Charleston distribution location. The individual products are displayed on the x axis and the number of case equivalents are displayed on the y axis (Figure 3.4). Figure 1 is an informative graphic because it shows how much of each product we currently have and which products need to be replenished. If the data does not have the column “Location” then it plots a bar graph of the second and third numeric columns in the data set.

Figure 2

For the second graphic, if the data that is being interpreted has columns called “Store-Count30days”, then it produces an overlaid bar graph of the number of retailers that have received Bogmayer product in the past 30, 60, and 90 days by product. This is an informative figure because it shows how much of each product we currently have and which need to be replenished. For the second visualization, the fallback graph produces a pie chart of the first string column and the first numeric column.

Insights

This section resulted in something different from what I imagined. My initial idea was to use an API key from a Large Language Model Platform to write insights and

Visualizations

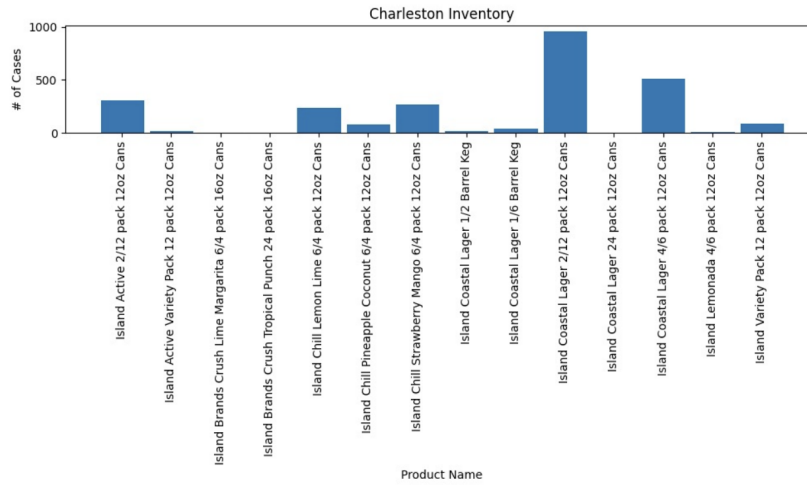


Figure 3.4: Charleston Inventory Visualization

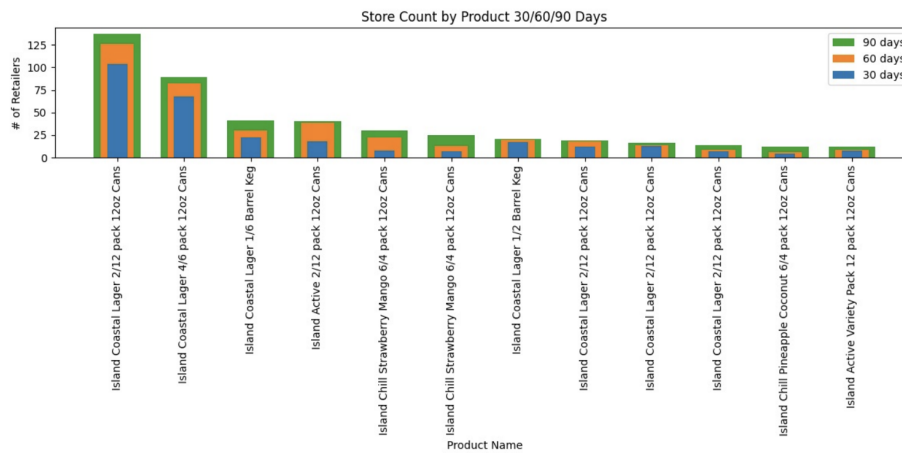


Figure 3.5: Comparison of Shipments

Summary Stats & Insights

The dataset contains 120 rows and 9 columns. There are 6 numeric columns and 3 categorical columns. It has 503 missing values in total. Numeric column 'On Floor Inventory (Cases)': min = 0.00, max = 8579.00, mean = 150.23. Numeric column 'Next Receive Units (Cases)': min = 10.00, max = 104.00, mean = 57.00. Numeric column 'On Floor Inventory (Case Equivs)': min = 0.00, max = 8808.81, mean = 154.60. Columns with most missing data: 'Next Receive Date' has 116 missing values. 'Next Receive Units (Cases)' has 116 missing values. Highly correlated numeric columns (>0.8): 'On Floor Inventory (Cases)' and 'On Floor Inventory (Case Equivs)' with correlation of 1.00. 'On Floor Inventory (Cases)' and 'StoreCount_30days' with correlation of 0.92. 'On Floor Inventory (Case Equivs)' and 'StoreCount_30days' with correlation of 0.93. 'StoreCount_30days' and 'StoreCount_60days' with correlation of 0.99. 'StoreCount_30days' and 'StoreCount_90days' with correlation of 0.98. 'StoreCount_60days' and 'StoreCount_90days' with correlation of 0.99.

Figure 3.6: Paragraph of Insights

important information about the data, the EDA, and the visualizations. This proved to be unsuccessful because the free version of OpenAI's API key [8] did not have the ability to handle that information. After trying to make this work for about ten days, I switched routes to a preset template that fills in information from the descriptive statistics. It does not provide any additional insights about the data, but it is helpful to see the results in written form. With this method, there will not be any AI hallucinations that will present faulty results. I used a Python script that is a template paragraph filled with f strings to be able to load in new information for every report that is uploaded.

Pipeline

Creating the pipeline was one of the most difficult tasks in my project. After researching how to automatically pull reports from Bogmayer's data platforms, Selenium presented to be the best successful option. I had to create this separate from my current dashboard because I needed to get this done quickly for my boss without messing up my current progress that I had made on this project. With Selenium in my Python script I automated the procedure to login and downloading current inventory reports and save them to my computer. This sounds a lot easier than it actually is because the data platforms have regulations and guidelines that I had to work around to be able to be successful. Once I had this functional locally on my computer, I put it into Github [4] to be able to run a Workflow Script to run it once per week regardless of whether my computer was on or not. Migrating my code into Github took me about ten days because I have needed to adjust my script in GitHub to include secrets rather than displaying confidential information, run Selenium with a headless driver, and create the workflow to send it to recipients on a designated

Location	Product Name	On Floor Inventory (Cases)	Next Receive Units (Cases)	Next Receive Date	On Floor Inventory (Case Equivs)	StoreCount_3 Days	StoreCount_6 Days	StoreCount_9 Days
Southern Crown Partners: Myrtle Beach, SC	Island Chill Pineapple Coconut 6/4 pack 12oz Cans	35.0	nan	nan	35.0	nan	nan	1.0
Southern Crown Partners: Myrtle Beach, SC	Island Chill Strawberry Mango 6/4 pack 12oz Cans	32.0	nan	nan	32.0	nan	4.0	5.0
Southern Crown Partners: Myrtle Beach, SC	Island Coastal Lager 1/2 Barrel Keg	5.0	nan	nan	34.44	5.0	5.0	5.0
Southern Crown Partners: Myrtle Beach, SC	Island Coastal Lager 1/6 Barrel Keg	8.0	nan	nan	18.37	8.0	11.0	12.0
Southern Crown Partners: Myrtle Beach, SC	Island Coastal Lager 2/12 pack 12oz Cans	39.0	nan	nan	39.0	13.0	15.0	18.0
Southern Crown Partners: Myrtle Beach, SC	Island Coastal Lager 24 pack 12oz Cans	0.0	nan	nan	0.0	nan	nan	nan
Southern Crown Partners: Myrtle Beach, SC	Island Coastal Lager 4/6 pack 12oz Cans	7.0	nan	nan	7.0	4.0	4.0	6.0
Southern Crown Partners: Myrtle Beach, SC	Island Lemonada 4/6 pack 12oz Cans	0.0	nan	nan	0.0	nan	nan	nan
Southern Crown Partners: Myrtle Beach, SC	Island Variety Pack 12 pack 12oz Cans	0.0	nan	nan	0.0	nan	1.0	1.0
Southern Crown Partners: Charleston, SC	Total	8852.0	204.0	11/19/2025	9104.78	nan	1.0	2.0
Southern Crown Partners: Charleston, SC	*Products Not mapped	6102.0	nan	nan	6102.0	nan	nan	nan
Southern Crown Partners: Charleston, SC	Island Active 2/12 pack 12oz Cans	378.0	nan	nan	378.0	18.0	25.0	27.0
Southern Crown Partners: Charleston, SC	Island Active Variety Pack 12 pack 12oz Cans	26.0	104.0	11/21/2025	26.0	8.0	9.0	13.0
Southern Crown Partners: Charleston, SC	Island Brands Crush Lime Margarita 6/4 pack 16oz Cans	6.0	nan	nan	8.0	nan	nan	nan
Southern Crown Partners: Charleston, SC	Island Brands Crush Tropical Punch 24 pack 16oz Cans	6.0	nan	nan	8.0	nan	nan	nan
Southern Crown Partners: Charleston, SC	Island Chill Lemon Lime 6/4 pack 12oz Cans	246.0	nan	nan	246.0	4.0	5.0	10.0
Southern Crown Partners: Charleston, SC	Island Chill Pineapple Coconut 6/4 pack 12oz Cans	79.0	nan	nan	79.0	4.0	7.0	13.0
Southern Crown Partners: Charleston, SC	Island Chill Strawberry Mango 6/4 pack 12oz Cans	119.0	nan	nan	119.0	8.0	21.0	33.0
Southern Crown Partners: Charleston, SC	Island Coastal Lager 1/2 Barrel Keg	33.0	nan	nan	227.33	17.0	20.0	22.0
Southern Crown Partners: Charleston, SC	Island Coastal Lager 1/6 Barrel Keg	42.0	100.0	11/19/2025	96.44	22.0	28.0	38.0
Southern Crown Partners: Charleston, SC	Island Coastal Lager 2/12 pack 12oz Cans	1101.0	nan	nan	1101.0	104.0	123.0	137.0

Figure 3.7: Inventory PDF

schedule. This stage came with frustration and many complications, but watching Youtube videos and reading documentation [13] helped me overcome these challenges and remain on my scheduled time frame.

Downloading Results

Downloading the results produced by the dashboard was also a difficult task that took me over two weeks to achieve. I tried multiple different packages: pdfkit [11] and PyPDF2 [12], and either nothing would happen when I clicked the download button or it would just create a blank document. After about ten days of no progress, I decided to create two different download buttons: a button that downloads the html file of the ydata report and a PDF file of all of the report. I wanted to do this because I could not get the PDF file to work, so I wanted some functionality in my download aspect of the program. I was able to successfully download the ydata report with a package called wkhtml in pdfkit [11], that exports the html file without losing any of the data. After this became functional, I moved onto the PDF section and eventually created a PDF report of the findings using PyPDF2 [12]. Even though I do not attach

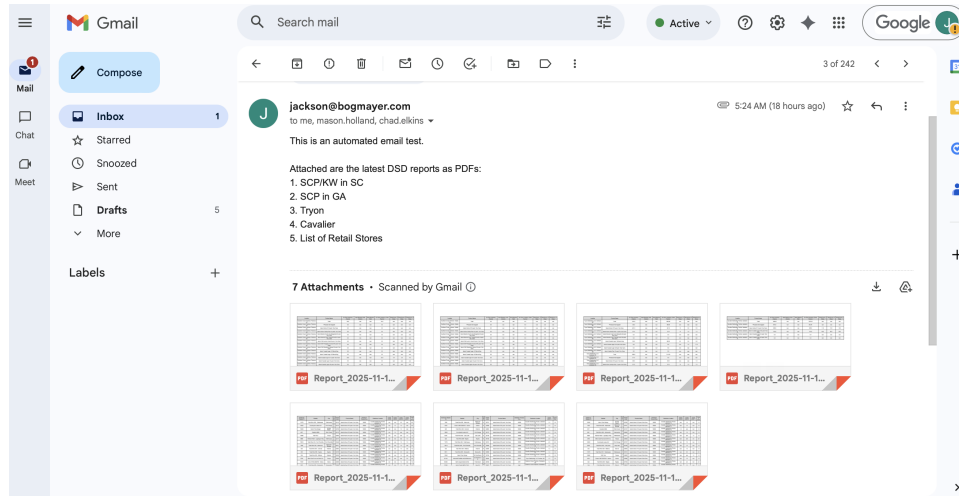


Figure 3.8: Picture of Files Attached to Email

the html file to the automated report, it is still an important feature if I want to go in and manually upload data and produce in depth, interactive results.

Email Automation

The email automation feature was pretty simple. After setting up a Google Developer account, I acquired a gmail API key [5] that I integrated into my code to send out emails. I wrote a function to attach different reports from my main and list who are the recipients and it sends an email out whenever this script is run. With the GitHub Workflows, I am able to download the reports from the data platforms, run the analytics dashboard on it, and send out all of the necessary emails with n number of attachments. This was one of the first things that I got working in my project because my boss needed me to send out inventory reports to him daily. Once I got this functional, I integrated my analytics portion of the project to email and automation feature. The automation of emailing is an integral part of my project because it allows me to be able to broadcast the results to the management of Bogmayer Ltd. The platform would not be helpful if the results do not get displayed and sent to the company.

Skills Used

Application of Previous Knowledge	Biggest Concepts Learned
HTML/CSS to build a website	GitHub Secrets, Workflows, and API key exploration
Pandas and Matplotlib for Analysis	JavaScript to expand cards, improve dashboard fluidity, and upload data frames
	Building a back end in Python
	download features (change)

Table 3.2: Application of Previous Knowledge vs. Biggest Concepts Learned

Complications

This was an experimental project where I encountered many problems and had to determine how to resolve them. The first complication I ran into was trying to login and download reports through my company's data platform using Selenium. I had to work around a lot of security checks and blockers to ensure that I could automatically log into my account, and then write a script to download pre-made reports to a folder without any time out errors. This whole process took around nine days to finally work and is an essential feature to my program. Secondly, I struggled to produce insights with an API key. Unfortunately, I was unable to resolve this issue and had to reroute my approach for the insights section. I switched to include a template paragraph filled with f strings and fill it in with the basic descriptive statistics every time it runs. This proved to be successful even though it was not my initial approach. Finally, the biggest problem I faced was close to the completion of my project. I struggled to download the Analysis report to a PDF and attach it to my automated email feature. This was frustrating because I could not find good services that are able to translate Python output and html file into pdfs and attach them to a consolidated report. This took me around three weeks to figure out completely, but PyPDF2 and wkhtmltopdf in pdfkit proved to be successful. After my Analysis report finally looked how I wanted it to look, it would not run through my automation process because it would time it out. After a lot of struggle, I simplified my report features and it finally was able to attach to my email and complete the functionality of my program.

Value to Bogmayer Ltd.

The Bogmayer Automated Analytics Dashboard successfully resolves the current data issues that existed in the company previously. The program accomplishes all of the goals that were set at the start of the semester. Sending out different customized reports and their analysis is extremely helpful for Bogmayer Ltd. allowing them to handle strategic decision making that is driven by facts. At the beginning of the project, I highlighted the goals and purpose of the program and it fully accomplishes all of the presented goals to drive progress in the company. The current reports that Bogmayer Ltd. is interested in consists of the inventory by distributor, sales, and the number of retailers selling each product. With the Automated Analytics Dashboard, we are downloading reports and running analysis on 4 different regions of distribution (South Carolina, Georgia, Florida, and North Carolina), and sending out the raw report data and the analysis reports. This program is currently running every morning at 6:30am Eastern Time to deliver impactful information to the executives of Bogmayer Ltd.

What I would have done differently

After completing the project, I realize that there were many things that I did well, but there were a few aspects that I wish I would have done differently during the project. First, my brainstorming and plan was very efficient, but I wish I had modeled my program after other Automated EDA platforms that I saw on GitHub and online. Mine is very unique, but I am worried that I am not doing things the optimal way. When brainstorming how to do my project, I looked at these other platforms for what they do, but I did not evaluate how they did it. Having the specific tools and packages layed out before hand would have been very useful, rather than working with a package and then finding out that it is not compatible with certain features or times out very quickly was very frustrating. Furthermore, I wish that I would have reached out to someone who has had experience with full-stack development for guidance. Having only one set of eyes is very restrictive because it is only one person's ideas and vision rather than communicating with somebody about what the most efficient way to complete something is.

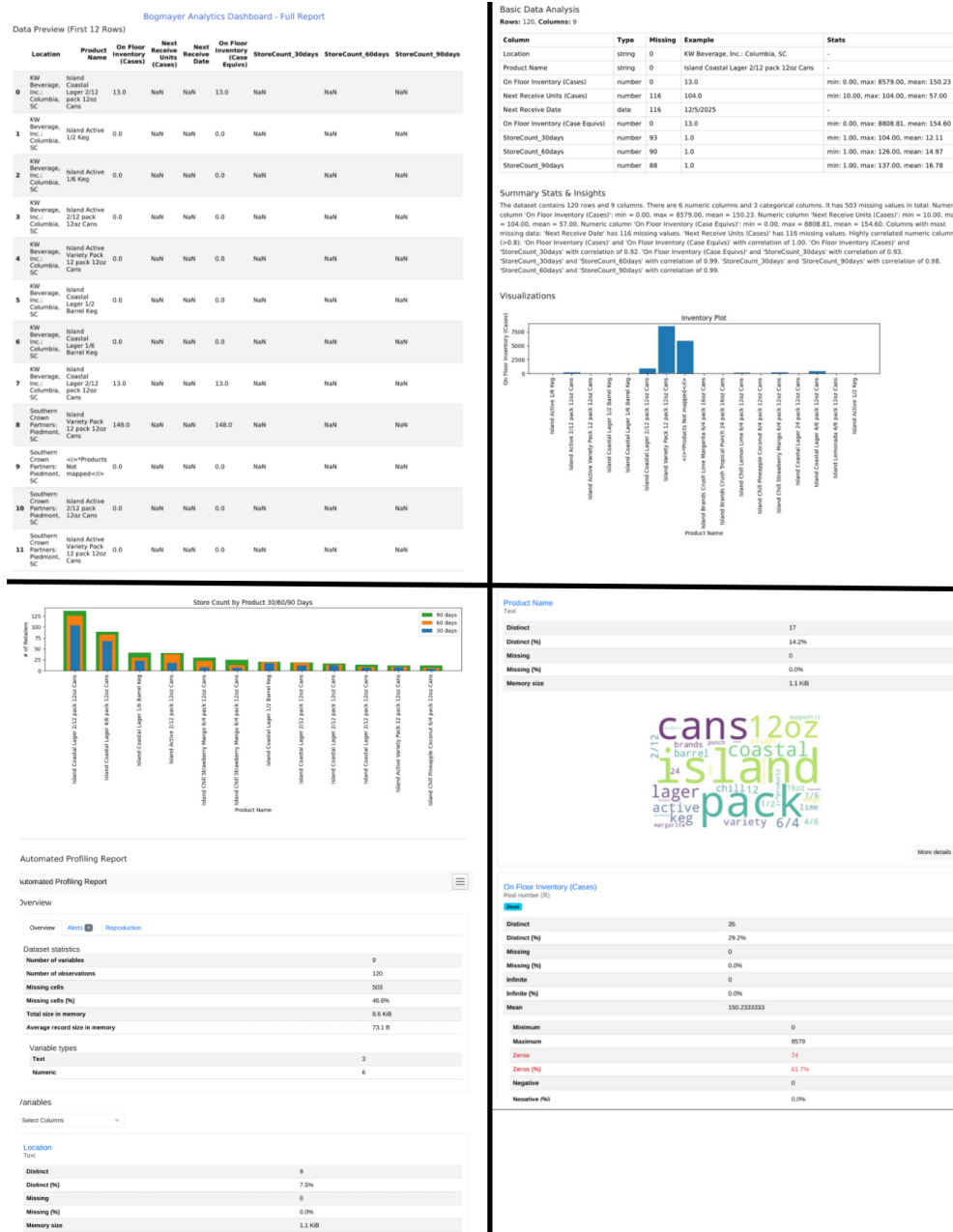


Figure 3.9: Pages 1-4 of Analytics Report

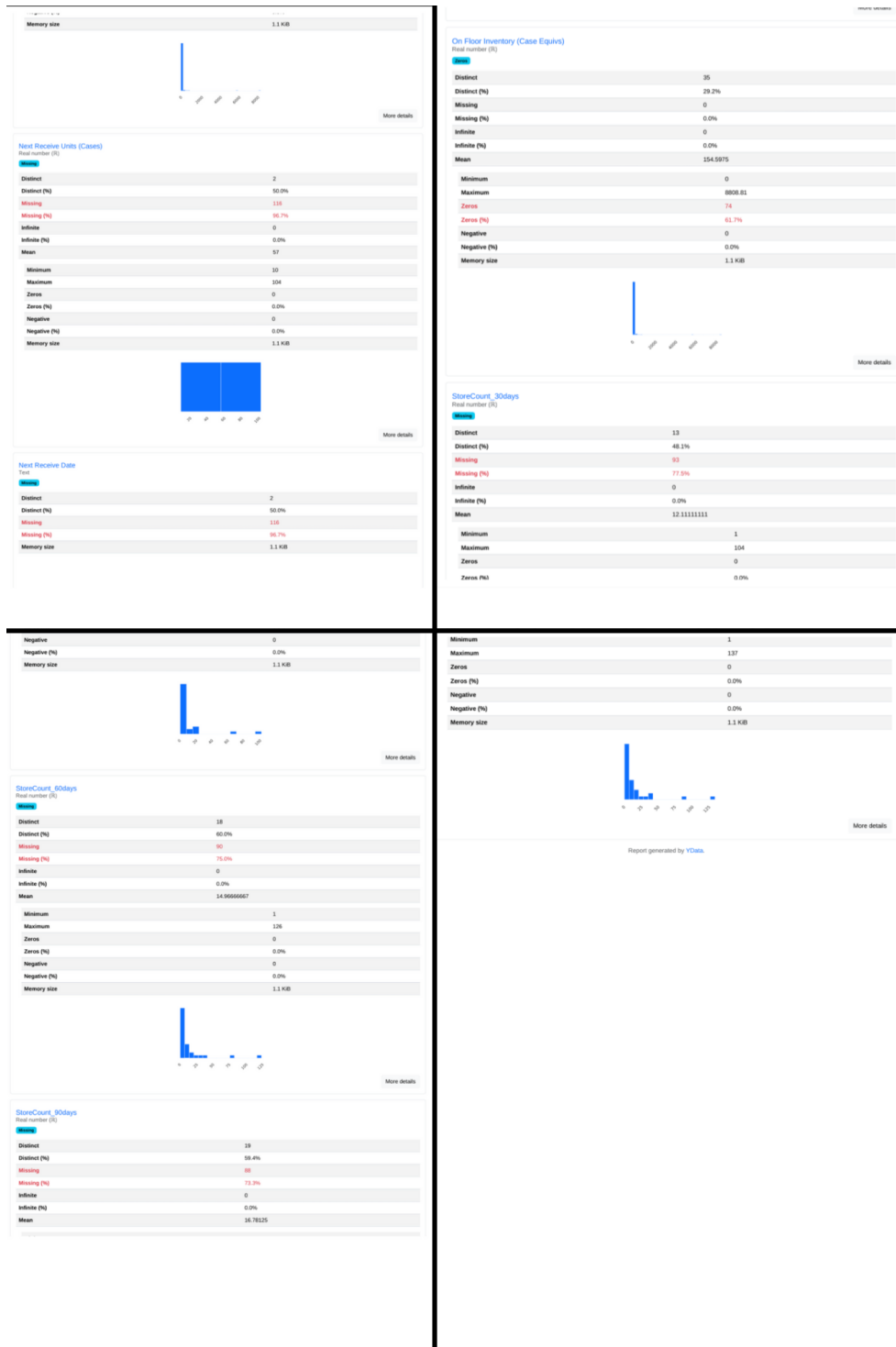


Figure 3.10: Pages 5-8 of Analytics Report

During the development of my project, my code was very loose and unstructured, which had both its benefits and complications. Having a mentor or someone I could reach out to would have been beneficial in all three stages of my project with giving me advice about planning, overcoming issues, and having a different perspective.

Future

Now that my program is complete, I will continue to advance the Bogmayer Automated Analytics dashboard after the semester. I am considering adding a series of adjustments to enhance the program's structure and functionality. My considerations include: storing all of the datasets into a centralized database that can be accessed from the onRender dashboard, adjusting the YData report to be personalized, adding encryption to the dashboard, and getting an AI API key to successfully analyze the data and produce insights. There are many adjustments and different ways to help Bogmayer Ltd. have interpretable results to bring success to the company. I am working with both the sales team and executives of Bogmayer Ltd. to evaluate the program and determine if there are any other reports I can attach or features that I can add before my internship ends.

Chapter 4

Conclusions

Reflection

Even though building my first functional, full-stack program came with many complications, I learned a lot about development, picked up new programming skills, and provided a solution to Bogmayer Ltd.'s current business issues. This task came with a lot of work put into it, perseverance, research, and learning new skills. This pressure is what led to my success of completing the project because I allocated eight hours of my time per week on this project, tried to complete my tasks before the deadline, and was driven to learn new things to get everything done. Something notable from working with a client is to build in extra time in your deadlines to be able to account for unexpected errors.

Skills Learned

Throughout the semester, I have learned both concrete technical skills, complimented by soft skills that will be critical in the work force. Technically, I learned a lot about how to set up a back end, the application of different python packages for specific uses, such as how to navigate the complications of a full stack project, converting python output to a pdf, and converting html to a pdf.

Feedback from Bogmayer Ltd.

When meeting with Bogmayer Ltd. to show them the final product, they were elated with what this can do for the company. Although a handful of employees in the company have already begun receiving the email reports, they were excited to see how it works and were thinking of the capabilities that it could bring to Bogmayer Ltd. My boss wants me to continue working on the platform to see if I can find additional ways to add impactful visualizations, store the data in a consolidated place, and be able to send out more reports every week.

Progression

After the semester ends, I plan to continue improving my program. Also, I plan on taking what I learned here and applying it to other scenarios, such as a different internship this spring, graduate schools, and my future job. This project covered an array of different topics that can translate into different roles in the future.

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